

Normal Forms

BSA, 2024

In[451]:=

```
ClearAll["Global`*"]
```

Input

In[452]:=

```
A = {{0, 1, 0, a}, {-1, 0, 0, 0}, {0, 0, 0, 1}, {0, 0, -1, 0}};  
B = A - λ * IdentityMatrix[4];  
A0 = A /. a → 0;  
A1 = A /. a → 1;
```

In[456]:=

```
Print["A = ", MatrixForm[A], ", A0 = ", MatrixForm[A0], ", A1 = ", MatrixForm[A1]]
```

$$A = \begin{pmatrix} 0 & 1 & 0 & a \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad A0 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad A1 = \begin{pmatrix} 0 & 1 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

In[457]:=

```
Print["B = ", MatrixForm[B]]
```

$$B = \begin{pmatrix} -\lambda & 1 & 0 & a \\ -1 & -\lambda & 0 & 0 \\ 0 & 0 & -\lambda & 1 \\ 0 & 0 & -1 & -\lambda \end{pmatrix}$$

Jordan form

In[458]:=

```
J0 = JordanDecomposition[A0][[2]];
J1 = JordanDecomposition[A1][[2]];

```

In[460]:=

```
Print["J0 = ", MatrixForm[J0], ", J1 = ", MatrixForm[J1]]

```

$$J_0 = \begin{pmatrix} -i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & i \end{pmatrix}, \quad J_1 = \begin{pmatrix} -i & 1 & 0 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & i & 1 \\ 0 & 0 & 0 & i \end{pmatrix}$$

Stability condition

In[461]:=

```
StabilityCondition[M_] := Module[{S, J}, {S, J} = JordanDecomposition[M];
  Length[Cases[Table[If[Re[J[[i, i]]] < 0 || (Re[J[[i, i]] == 0 && Re[J[[i, i + 1]] == 0), 1, 0], {i, 1, Length[J] - 1}], x_ /; x == 0]] == 0 &&
  Re[J[[-1, -1]]] ≤ 0
];
CheckStability[M_] := If[StabilityCondition[M], Print["Stable"], Print["Unstable"]];

```

In[463]:=

```
CheckStability[A0]

```

Stable

In[464]:=

```
CheckStability[A1]

```

Unstable

Smith canonical form

In[465]:=

```
DD[n_, M_] := PolynomialGCD @@ Flatten[Minors[M, n]];
F[n_, M_] := Factor[Expand[DD[n, M] / DD[n - 1, M]]];
S0 = DiagonalMatrix[Table[F[k, B /. a -> 0], {k, 4, 1, -1}]];
S1 = DiagonalMatrix[Table[F[k, B /. a -> 1], {k, 4, 1, -1}]];
```

In[469]:=

```
Print["S0 = ", MatrixForm[S0], ", S1 = ", MatrixForm[S1]]
```

$$S_0 = \begin{pmatrix} 1 + \lambda^2 & 0 & 0 & 0 \\ 0 & 1 + \lambda^2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, S_1 = \begin{pmatrix} (1 + \lambda^2)^2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Compute the solution

In[470]:=

```
X = {x1[t], x2[t], x3[t], x4[t]};
InX = {x1[0] == 1, x2[0] == 1, x3[0] == 1, x4[0] == 1};
Eq0 = Thread[D[X, t] == A0.X];
Eq1 = Thread[D[X, t] == A1.X];
Sol0 = X /. DSolve[Union[Eq0, InX], X, t][[1]];
Sol1 = X /. DSolve[Union[Eq1, InX], X, t][[1]];
```

In[476]:=

```
Print["X0 = ", MatrixForm[Sol0], ", X1 = ", MatrixForm[Sol1]]
```

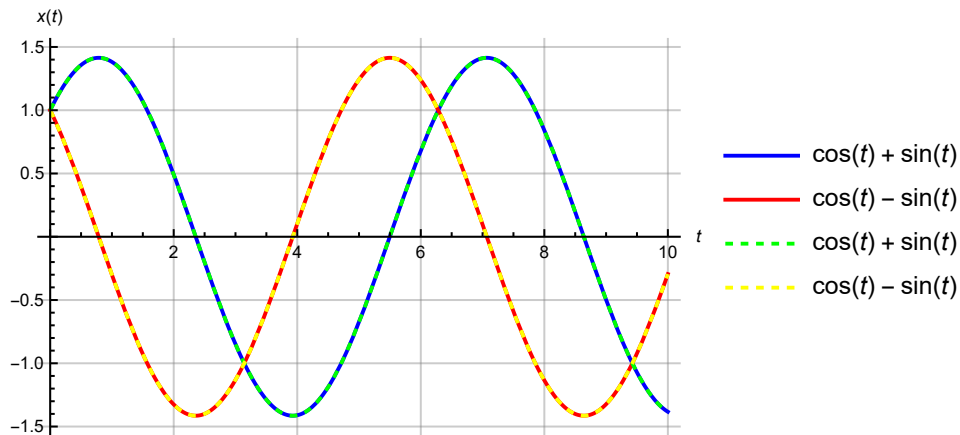
$$X0 = \begin{pmatrix} \cos[t] + \sin[t] \\ \cos[t] - \sin[t] \\ \cos[t] + \sin[t] \\ \cos[t] - \sin[t] \end{pmatrix}, X1 = \begin{pmatrix} \frac{1}{2} (2 \cos[t] + t \cos[t] + 3 \sin[t] - t \sin[t]) \\ \frac{1}{2} (2 \cos[t] - t \cos[t] - \sin[t] - t \sin[t]) \\ \cos[t] + \sin[t] \\ \cos[t] - \sin[t] \end{pmatrix}$$

Plot the solution

In[477]:=

```
Plot[Sol0, {t, 0, 10}, PlotLegends -> Placed["Expressions", Right], AxesLabel -> {t, x[t]},  
GridLines -> Automatic, PlotStyle -> {Blue, Red, {Green, Dashed}, {Yellow, Dashed}}, PlotRange -> All]
```

Out[477]=



In[478]:=

```
Plot[Sol1, {t, 0, 10}, PlotLegends → Placed["Expressions", Right], AxesLabel → {t, x[t]},
  GridLines → Automatic, PlotStyle → {Blue, Red, Green, Orange}, PlotRange → All]
```

Out[478]=

